

Clustering

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Clustering_2
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* K-Means:-

Goal:- Minimize the distance between the points within a cluster. (wrt their centroid)

→ k-means is a centroid based / distance-based algorithm.

Step 1:-

Choose the number of clusters k.

Step 2:-

Select k random points from the data as centroids.

Step 3:- Assign all the points to the closest cluster centroid

Step 4:- Re-compute the centroids of newly formed clusters:- centroids = average of all points in cluster.

Step 5:- Repeat step 3 & 4, until one of the stopping criteria doesn't hit.

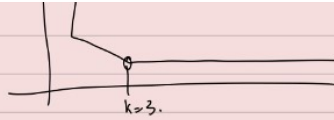
- 1) Centroids of the newly formed cluster doesn't change.
- 2) Points remain in the same cluster.
- 3) Maximum number of iterations are reached.

○ How to choose optimal k?

1) Elbow Method:-

→ find WSS (Within-cluster-sum of squared errors) for different values of k, and choose the k for which WSS becomes first starts to diminish

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$$WSS = \sum_{k=1}^K \sum_{i \in S_k} \sum_{j=1}^P (x_{ij} - \bar{x}_{kj})^2$$

Sum over all cluster
 over all points in K.
 centroid of cluster K
 square Euclidean distance over the entire dimension

2) The Silhouette Method.

⇒ The Silhouette Score measures how similar a point is to its own cluster (cohesion) compared to other clusters.

↳ range of silhouette values are between +1, and -1.

↳ High value is desirable, and indicates that point is placed in correct cluster.

→ If many points have -ve silhouette score, it may indicate that

we have created too many, or too few clusters.

Silhouette Score :-

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \text{ if } |C_i| > 1.$$

$$\text{and } s(i) = 0, \text{ if } |C_i| = 1$$

⇒ $s(i) = 0$ if i is the only point in the cluster.

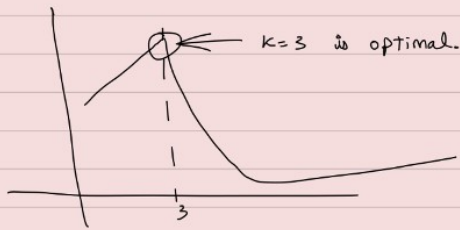
⇒ $a(i)$ is the measure of the similarity of point i , with its own cluster. It's measured as the average distance of i from other points in the same cluster as i .

$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, j \neq i} d(i, j)$$

Similarly, $b(i)$ is the measure of dissimilarity of i from points in other clusters.

$$b(i) = \min_{j \neq i} \frac{1}{|C_j|} \sum_{j \in C_j} d(i, j).$$

from sklearn.metrics import silhouette_score



3) Dunn - Index :- (Dis-similarity between the clusters)

$$\text{Dunn Index} = \frac{\min(\text{Inter cluster distance})}{\max(\text{Intra cluster distance})}$$

* Assumption of K-means :-

1) As K-means calculates cluster distance from centroid it forms a spherical shape clusters, thus it cannot cluster complicated geometrical shapes.

↳ Solution :- Kernel Method.

2) Size of clusters :-

↳ This algorithm considers only distance, thus does not account for clusters with different sizes, or densities.

3) It assumes features within a cluster have equal variance.

4) Requires good initialization.

* K-Means ++ :-

→ Overcomes the random initialization problem.

→ Exactly same as K-means, just has smarter initialization.

* Steps:-

1) Randomly select the first centroid from the data points.

2) For each data point compute its distance from the nearest, previously chosen centroid.

3) Select the next centroid from the data points such that the probability of choosing a point as centroid is directly proportional to its distance from the nearest, previously chosen centroid.

5. Also one more disadvantage is the use of Euclidean distance.

↳ i.e. choose the point having maximum distance from the last chosen centroid.

4) Repeat 2 & 3 until k centroids have been sampled.

Q) What happens when features don't have same scale & variance?

⇒ when features have different scales or variances, k-mean clustering tends to become dominated by the features with larger numerical ranges.

⇒ If one feature has values ranging in thousands while another varies between 0 & 10, the distance metric will be largely influenced by thousands-range feature.

↳ It can lead to clusters that reflect differences in that one feature rather than a balanced combination of all features.

ex: Income vs Age

⇒ Imagine a dataset where one feature is annual income (20k\$ to 200k\$) & another is age (20 to 70 years).

↳ Without Normalization, income differences will dominate the distance calculations.

• How to fix?

⇒ The common solution is to standardize or normalize the data.

Q) What if scale is different, but variance is same?

⇒ the euclidean distance formula has the following formula:

$$\text{Distance} = \sqrt{\underbrace{(x - \mu_x)^2}_{\text{distance of } x \text{ in } \mu_x} + \underbrace{(y - \mu_y)^2}_{\text{distance of } y \text{ in } \mu_y}}$$

Variance for feature x $= s^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2$

You can ignore the term as this is same across all features (number of data points are considered same across all features)

$\therefore$ If variances are same then the contribution of each feature is same, hence euclidean distance which has a weight of 2 is justified and unbiased.